BS-M301 — Probability and Statistics

Semester III • 3L:0T:3P • 3 Credits

Module 3 Exam Notes: Bivariate Distributions, Sums, Quotients and Bayes' Rule

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Bivariate Distributions and Their Fundamental Properties

In numerous engineering problems, physical phenomena involve the simultaneous observation of two interdependent random quantities, such as the in-phase and quadrature components of an RF communication signal (I, Q), transmitter voltage and received current (V, I), or temperature and humidity. These are modeled jointly as a **two-dimensional (bivariate) random variable** $(X, Y) : \Omega \to \mathbb{R}^2$.

Definition: Joint Probability Mass and Density Functions

Discrete Bivariate Case (Joint PMF): Let (X, Y) take values in a countable set of pairs (x_i, y_j). The **Joint PMF** is:

$$p_{XY}(x_i, y_j) = P(X = x_i, Y = y_j) \tag{1}$$

Axioms: (1) $p_{XY}(x_i, y_j) \geq 0, \quad \forall (i, j),$ (2) $\sum_i \sum_j p_{XY}(x_i, y_j) = 1.$

Continuous Bivariate Case (Joint PDF): (X, Y) is jointly continuous if there exists an integrable surface $f_{XY} : \mathbb{R}^2 \to \mathbb{R}$ such that for any two-dimensional region $D \subseteq \mathbb{R}^2$:

$$P((X, Y) \in D) = \iint_D f_{XY}(x, y)\, dxdy \tag{2}$$

Axioms:

$$\textbf{Non-negativity:} \quad f_{XY}(x, y) \geq 0, \quad \forall (x, y) \in \mathbb{R}^2 \tag{3}$$

$$\textbf{Unit Volume:} \quad \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y)\, dxdy = 1 \tag{4}$$

1.1 Joint Cumulative Distribution Function (Joint CDF)

The **Joint CDF** of (X, Y) is defined as the probability of the infinite quadrant to the south-west of (x, y):

$$F_{XY}(x, y) = P(X \leq x, Y \leq y) = \begin{cases} \sum_{x_i \leq x} \sum_{y_j \leq y} p_{XY}(x_i, y_j), & \text{(Discrete)} \\ \int_{-\infty}^{x} \int_{-\infty}^{y} f_{XY}(u, v)\, dvdu, & \text{(Continuous)} \end{cases} \tag{5}$$

Properties of the Joint CDF and Recovery of Joint PDF

1. **Boundary Limits:**

$$F_{XY}(-\infty, y) = 0, \qquad F_{XY}(x, -\infty) = 0, \qquad F_{XY}(-\infty, -\infty) = 0 \tag{6}$$

$$F_{XY}(+\infty, +\infty) = 1 \tag{7}$$

2. **Marginal Recovery from Joint CDF:**

$$F_X(x) = \lim_{y \to +\infty} F_{XY}(x, y) = F_{XY}(x, \infty), \qquad F_Y(y) = \lim_{x \to +\infty} F_{XY}(x, y) = F_{XY}(\infty, y) \tag{8}$$

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3. **Recovery of Continuous Joint PDF by Partial Differentiation:** At every continuity point of $f_{XY}(x, y)$:

$$f_{XY}(x, y) = \frac{\partial^2}{\partial x \partial y} F_{XY}(x, y) = \frac{\partial^2}{\partial y \partial x} F_{XY}(x, y) \tag{9}$$

4. **Rectangle Probability Rule:** For any rectangular region $[x_1, x_2] \times [y_1, y_2]$:

$$P(x_1 < X \leq x_2,\ y_1 < Y \leq y_2) = F_{XY}(x_2, y_2) - F_{XY}(x_1, y_2) - F_{XY}(x_2, y_1) + F_{XY}(x_1, y_1) \geq 0 \tag{10}$$

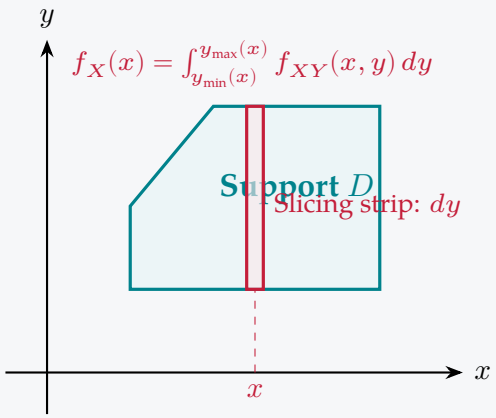
1.2 Marginal Distributions

The individual distribution of one variable obtained by integrating or summing out the other variable over its entire support is termed the **marginal distribution**.

Marginal PMFs: $$p_X(x_i) = \sum_j p_{XY}(x_i, y_j), \qquad p_Y(y_j) = \sum_i p_{XY}(x_i, y_j) \tag{11}$$

Marginal PDFs: $$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y)\, dy, \qquad f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y)\, dx \tag{12}$$

TikZ Schematic: Geometric Interpretation of Marginal Integration across Joint Support



Area under the cross-sectional slice at fixed x gives the marginal density $f_X(x)$.

2. Conditional Distributions, Densities and Bayes' Rule

Definition: Conditional Probability Density Function

Given that the random variable Y has taken on a specific numerical value $Y = y$ (where marginal density $f_Y(y) > 0$), the **conditional PDF** of X given $Y = y$ is defined as:

$$f_{X|Y}(x \mid y) = \frac{f_{XY}(x, y)}{f_Y(y)} = \frac{f_{XY}(x, y)}{\displaystyle\int_{-\infty}^{\infty} f_{XY}(u, y)\, du} \tag{13}$$

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Symmetrically, the conditional PDF of Y given $X = x$ (with $f_X(x) > 0$) is:

$$f_{Y|X}(y \mid x) = \frac{f_{XY}(x,y)}{f_X(x)} = \frac{f_{XY}(x,y)}{\int_{-\infty}^{\infty} f_{XY}(x,v)\,dv} \tag{14}$$

Every conditional density satisfies the standard PDF axioms: $f_{X|Y}(x \mid y) \geq 0$ and $\int_{-\infty}^{\infty} f_{X|Y}(x \mid y)dx = 1$.

2.1 Conditional Expectation and the Tower Property

- **Conditional Expectation:** The mean of X given $Y = y$ is a function of y:

$$E[X \mid Y = y] = \int_{-\infty}^{\infty} x f_{X|Y}(x \mid y)\,dx \tag{15}$$

- **Law of Iterated Expectations (Tower Property):**

$$E\left[E[X \mid Y]\right] = E[X] \tag{16}$$

Proof:

$$\begin{aligned} E\left[E[X \mid Y]\right] &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} x \frac{f_{XY}(x,y)}{f_Y(y)} dx \right) f_Y(y)\,dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{XY}(x,y)\,dxdy = \int_{-\infty}^{\infty} x \left(\int_{-\infty}^{\infty} f_{XY}(x,y)dy \right) dx \\ &= \int_{-\infty}^{\infty} x f_X(x)\,dx = E[X] \end{aligned} \tag{17}$$

- **Law of Total Variance (Eve's Law):**

$$\mathrm{Var}(X) = E[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(E[X \mid Y]) \tag{18}$$

Total uncertainty equals the average unexplained variance plus the variance explained by Y.

2.2 Bayes' Rule for Continuous and Mixed Random Variables

Continuous and Mixed Formulations of Bayes' Rule

Continuous Inversion (Posterior Density): Equating $f_{XY}(x,y) = f_X(x)f_{Y|X}(y \mid x) = f_Y(y)f_{X|Y}(x \mid y)$:

$$f_{X|Y}(x \mid y) = \frac{f_X(x) \cdot f_{Y|X}(y \mid x)}{f_Y(y)} = \frac{f_X(x) \cdot f_{Y|X}(y \mid x)}{\int_{-\infty}^{\infty} f_X(u) \cdot f_{Y|X}(y \mid u)\,du} \tag{19}$$

where $f_X(x)$ is the *prior density*, $f_{Y|X}(y \mid x)$ is the *likelihood function*, and $f_{X|Y}(x \mid y)$ is the *posterior density*.

Discrete Hypothesis with Continuous Measurement (Signal Detection): Let

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$\{H_1, H_2, \ldots, H_n\}$ be discrete transmitted symbols (priors $P(H_i)$) observed through continuous additive noise Y:

$$P(H_k \mid Y = y) = \frac{P(H_k) \cdot f_{Y|H_k}(y)}{\sum_{i=1}^{n} P(H_i) \cdot f_{Y|H_i}(y)} \tag{20}$$

3. Bivariate Transformations and the Jacobian Method

When analyzing signal processing systems, inputs (X, Y) are mapped into new output variables (U, V) via mathematical functions $U = g_1(X, Y)$ and $V = g_2(X, Y)$.

Theorem: Bivariate Jacobian Transformation

Let (X, Y) have joint PDF $f_{XY}(x, y)$ supported on set D. Let the transformation:

$$u = g_1(x, y), \qquad v = g_2(x, y) \tag{21}$$

define a one-to-one and invertible mapping from D onto range S, with unique inverse:

$$x = h_1(u, v), \qquad y = h_2(u, v) \tag{22}$$

If the partial derivatives are continuous, the **Jacobian determinant** J of the inverse transformation is:

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u} \tag{23}$$

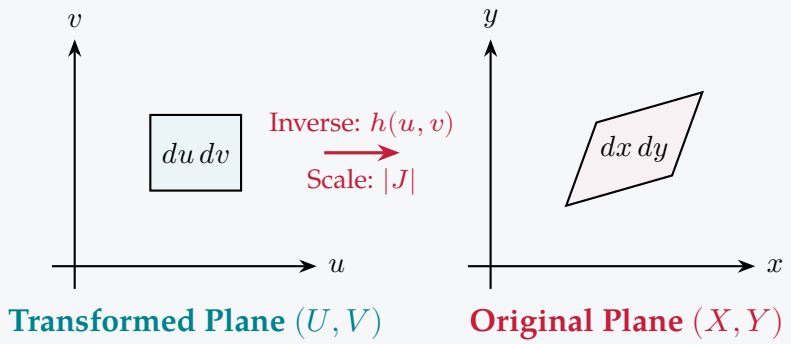
Then the joint PDF of (U, V) over S is given by:

$$f_{UV}(u, v) = f_{XY}\left(h_1(u, v), h_2(u, v)\right) \cdot |J| \tag{24}$$

where $|J|$ is the absolute value of the Jacobian determinant (representing the local area scaling factor: $dxdy = |J|dudv$).

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TikZ Schematic: Area Element Transformation under Jacobian Mapping



4. Distributions of Sums and Quotients

4.1 Distribution of the Sum $Z = X + Y$ (Convolution)

Derivation: PDF of the Sum of Two Continuous Random Variables

To find the PDF of $Z = X + Y$, introduce an auxiliary variable $W = X$. The forward transformation is:

$$z = x + y, \qquad w = x \tag{25}$$

The unique inverse transformation is:

$$x = w, \qquad y = z - w \tag{26}$$

Computing the Jacobian determinant:

$$J = \det \begin{pmatrix} \frac{\partial x}{\partial z} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial z} & \frac{\partial y}{\partial w} \end{pmatrix} = \det \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix} = (0)(-1) - (1)(1) = -1 \implies |J| = |-1| = 1 \tag{27}$$

The joint PDF of (Z, W) is:

$$f_{ZW}(z, w) = f_{XY}(w, z - w) \cdot |J| = f_{XY}(w, z - w) \tag{28}$$

Integrating out the auxiliary variable w yields the marginal PDF of the sum Z:

$$f_Z(z) = \int_{-\infty}^{\infty} f_{ZW}(z, w)\, dw = \int_{-\infty}^{\infty} f_{XY}(w, z - w)\, dw \tag{29}$$

If X and Y are **independent** ($f_{XY}(x, y) = f_X(x) f_Y(y)$), this becomes the fundamental **continuous convolution integral**:

$$f_Z(z) = (f_X * f_Y)(z) = \int_{-\infty}^{\infty} f_X(w) f_Y(z - w)\, dw = \int_{-\infty}^{\infty} f_X(z - y) f_Y(y)\, dy \tag{30}$$

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4.2 Distribution of the Quotient $Z = X/Y$

Derivation: PDF of the Quotient of Two Random Variables

Let $Z = \frac{X}{Y}$. Introduce the auxiliary variable $W = Y$. The forward transformation is:

$$z = \frac{x}{y}, \qquad w = y \quad (w \neq 0) \tag{31}$$

The inverse transformation is:

$$x = z \cdot w, \qquad y = w \tag{32}$$

The Jacobian determinant is:

$$J = \det \begin{pmatrix} \frac{\partial x}{\partial z} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial z} & \frac{\partial y}{\partial w} \end{pmatrix} = \det \begin{pmatrix} w & z \\ 0 & 1 \end{pmatrix} = (w)(1) - (z)(0) = w \implies |J| = |w| \tag{33}$$

The joint PDF of (Z, W) is:

$$f_{ZW}(z, w) = f_{XY}(zw, w) \cdot |w| \tag{34}$$

Integrating out w over $(-\infty, \infty)$ yields the general quotient density:

$$f_Z(z) = \int_{-\infty}^{\infty} |w| f_{XY}(zw, w)\, dw \tag{35}$$

If X and Y are independent:

$$f_Z(z) = \int_{-\infty}^{\infty} |w| f_X(zw) f_Y(w)\, dw \tag{36}$$

4.3 Quotient of Two Independent Standard Normals: The Cauchy Distribution

Classic Result: Ratio of Independent Standard Normals follows Cauchy

Let $X \sim \mathcal{N}(0, 1)$ and $Y \sim \mathcal{N}(0, 1)$ be independent. Their joint density is:

$$f_{XY}(x, y) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \cdot \frac{1}{\sqrt{2\pi}} e^{-y^2/2} = \frac{1}{2\pi} e^{-(x^2+y^2)/2} \tag{37}$$

Applying the quotient formula to $Z = X/Y$:

$$f_Z(z) = \int_{-\infty}^{\infty} |w| \cdot \frac{1}{2\pi} e^{-(z^2 w^2 + w^2)/2}\, dw = \frac{1}{2\pi} \int_{-\infty}^{\infty} |w| e^{-\frac{w^2(1+z^2)}{2}}\, dw \tag{38}$$

Since the integrand is even with respect to w:

$$f_Z(z) = \frac{2}{2\pi} \int_0^{\infty} w e^{-\frac{w^2(1+z^2)}{2}}\, dw = \frac{1}{\pi} \int_0^{\infty} w e^{-\frac{w^2(1+z^2)}{2}}\, dw \tag{39}$$

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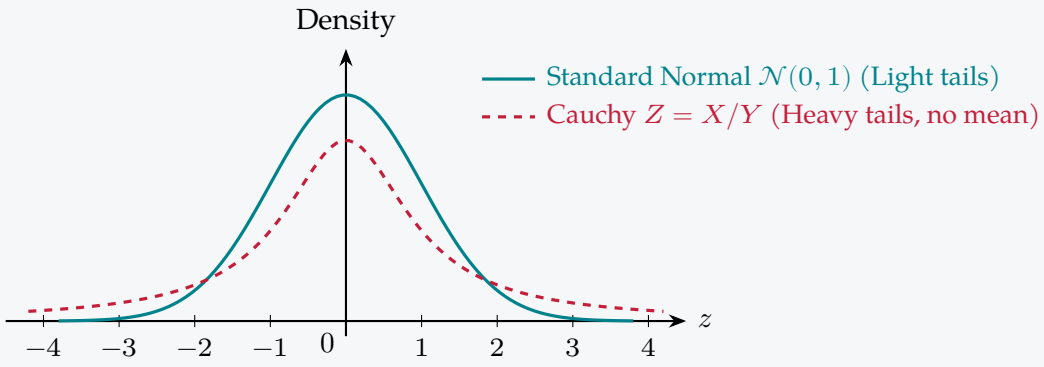
Substitute $u = \frac{w^2(1+z^2)}{2} \implies du = (1+z^2)wdw \implies wdw = \frac{du}{1+z^2}$:

$$f_Z(z) = \frac{1}{\pi(1+z^2)} \int_0^\infty e^{-u}\, du = \frac{1}{\pi(1+z^2)} \left[-e^{-u}\right]_0^\infty = \frac{1}{\pi(1+z^2)}(1) \tag{40}$$

$$f_Z(z) = \frac{1}{\pi(1+z^2)}, \qquad -\infty < z < \infty \tag{41}$$

This is the celebrated **Standard Cauchy Distribution**! *Crucial Exam Note:* For a Cauchy distribution, $\int_{-\infty}^{\infty} |z| \frac{1}{\pi(1+z^2)} dz = \infty$. Therefore, the Cauchy distribution has **no defined mean, no variance, and no MGF**!

TikZ Plot: Standard Cauchy PDF vs. Standard Normal Gaussian PDF



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Quick Revision — Module 3

Bivariate Distributions, Sums, Quotients and Bayes' Rule | BS-M301

Key Formulas and Bivariate Relations at a Glance

Continuous Joint PDF	$f_{XY}(x,y) \geq 0, \quad \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{XY}(x,y)dxdy = 1$
Joint CDF Formula	$F_{XY}(x,y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f_{XY}(u,v)dvdu, \quad f_{XY}(x,y) = \frac{\partial^2 F_{XY}}{\partial x \partial y}$
Marginal Densities	$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y)dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x,y)dx$
Conditional Densities	$f_{X\|Y}(x \mid y) = \frac{f_{XY}(x,y)}{f_Y(y)}, \quad f_{Y\|X}(y \mid x) = \frac{f_{XY}(x,y)}{f_X(x)}$
Statistical Independence	$f_{XY}(x,y) = f_X(x)f_Y(y) \iff f_{X\|Y}(x \mid y) = f_X(x)$
Tower Property	$E\left[E[X \mid Y]\right] = E[X], \quad \text{Var}(X) = E[\text{Var}(X\|Y)] + \text{Var}(E[X\|Y])$
Jacobian Transform	$f_{UV}(u,v) = f_{XY}(h_1,h_2)\cdot\|J\|, \quad J = \det\begin{pmatrix}\frac{\partial x}{\partial u} & \frac{\partial x}{\partial v}\\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v}\end{pmatrix}$
Sum PDF (Convolution)	$Z = X+Y \implies f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(w, z-w)dw$
Quotient PDF	$Z = X/Y \implies f_Z(z) = \int_{-\infty}^{\infty} \|w\| f_{XY}(zw, w)dw$
Cauchy from Normals	$X, Y \sim \mathcal{N}(0,1)$ indep. $\implies Z = \frac{X}{Y} \sim$ Cauchy : $f_Z(z) = \frac{1}{\pi(1+z^2)}$
Continuous Bayes' Rule	$f_{X\|Y}(x \mid y) = \frac{f_X(x)f_{Y\|X}(y\|x)}{\int_{-\infty}^{\infty} f_X(u)f_{Y\|X}(y\|u)du}$

One-Line Definitions

- **Joint PDF $f_{XY}(x,y)$:** Surface whose double integral over any two-dimensional region D gives $P((X,Y) \in D)$.
- **Marginal PDF:** Individual density of one component obtained by integrating out the other variable over $\mathbb{R}$.
- **Conditional PDF:** Sliced cross-sectional density of X normalized by $f_Y(y)$ under the condition that $Y = y$.
- **Statistical Independence:** Complete factorization $f_{XY}(x,y) = f_X(x)f_Y(y)$ across the entire domain.
- **Tower Property:** Fundamental law asserting that the overall expectation equals the expectation of conditional expectations.
- **Jacobian Determinant (J):** Local area transformation ratio $dxdy = |J|dudv$ relating transformed coordinate spaces.
- **Convolution Theorem:** Operation yielding the PDF of the sum of independent random variables: $f_Z = f_X * f_Y$.
- **Quotient Distribution:** Density $f_Z(z) = \int |w| f(zw, w)dw$ describing the ratio of two random variables.
- **Cauchy Distribution:** Heavy-tailed distribution $f(z) = \frac{1}{\pi(1+z^2)}$ resulting from $\mathcal{N}(0,1)/\mathcal{N}(0,1)$, lacking mean and variance.
- **Posterior Density:** Updated probability density of an unknown parameter after incorporating experimental observation y.

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Frequently Asked / Viva Questions (Part 1)

Q1. State the necessary and sufficient conditions for two continuous random variables X and Y to be independent.

X and Y are independent if and only if their joint PDF factors into the product of their marginal PDFs everywhere: $f_{XY}(x,y) = f_X(x)f_Y(y)$ for all $(x,y) \in \mathbb{R}^2$, and the support region D is a product of intervals (non-coupled boundaries).

Q2. How is the joint PDF obtained from the joint Cumulative Distribution Function?

By mixed second-order partial differentiation: $f_{XY}(x,y) = \frac{\partial^2}{\partial x \partial y} F_{XY}(x,y)$, provided the second partial derivatives exist and are continuous.

Q3. State and prove the Law of Iterated Expectations (Tower Property).

$E\left[E[X \mid Y]\right] = E[X]$. Proof: $E[E[X \mid Y]] = \int \left(\int x \frac{f(x,y)}{f_Y(y)} dx \right) f_Y(y) dy = \iint x f(x,y) dx dy = \int x f_X(x) dx = E[X]$.

Q4. What is the physical significance of the Jacobian determinant in bivariate transformations?

The Jacobian determinant $J = \frac{\partial(x,y)}{\partial(u,v)}$ represents the local geometric dilation factor between the original and transformed coordinate planes, ensuring conservation of probability mass: $P(dxdy) = f_{XY}(x,y)dxdy = f_{XY}(h_1,h_2)|J|dudv = f_{UV}(u,v)dudv$.

Q5. Derive the PDF of the sum $Z = X + Y$ of two independent continuous random variables.

Let $Z = X + Y, W = X \implies X = W, Y = Z - W$. The Jacobian is $|J| = 1$. The joint density is $f_{ZW}(z,w) = f_{XY}(w, z-w) = f_X(w)f_Y(z-w)$. Integrating over w gives the convolution: $f_Z(z) = \int_{-\infty}^{\infty} f_X(w)f_Y(z-w)dw$.

Q6. Why does the absolute value $|w|$ appear in the formula for the quotient density $Z = X/Y$?

The Jacobian of the inverse transformation $x = zw, y = w$ is $J = \det \begin{pmatrix} w & z \\ 0 & 1 \end{pmatrix} = w$.

Because an area element must always be strictly non-negative ($dxdy = |J|dzdw$), the absolute value $|J| = |w|$ must be taken.

Q7. Show that the ratio of two independent standard normal random variables follows a Cauchy distribution.

$f_Z(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} |w| e^{-w^2(1+z^2)/2} dw = \frac{1}{\pi} \int_0^{\infty} w e^{-w^2(1+z^2)/2} dw$. Substituting $u = w^2(1+z^2)/2$ transforms the integral to $\frac{1}{\pi(1+z^2)} \int_0^{\infty} e^{-u} du = \frac{1}{\pi(1+z^2)}$, which is the standard Cauchy PDF.

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Frequently Asked / Viva Questions (Part 2)

Q8. Why does the Cauchy distribution have no mathematical expectation or variance?

$E[|Z|] = \frac{1}{\pi}\int_{-\infty}^{\infty}\frac{|z|}{1+z^2}dz = \frac{2}{\pi}\int_0^{\infty}\frac{z}{1+z^2}dz = \frac{1}{\pi}\left[\ln(1+z^2)\right]_0^{\infty} = \infty$. Because the absolute integral diverges logarithmically due to heavy polynomial tails ($\sim 1/z^2$), the mean is mathematically undefined, and all higher moments diverge.

Q9. State the Law of Total Variance (Eve's Law) and interpret its terms.

$\mathrm{Var}(X) = E[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(E[X \mid Y])$. The first term represents the average unexplained variation remaining in X after conditioning on Y; the second term represents the variation in the conditional predictor $E[X \mid Y]$ explained by Y.

Q10. State Continuous Bayes' Theorem and explain its role in signal estimation.

$f_{X|Y}(x \mid y) = \frac{f_X(x)f_{Y|X}(y|x)}{\int f_X(u)f_{Y|X}(y|u)du}$. In communications, X is the transmitted signal (prior), $f_{Y|X}(y \mid x)$ is the channel noise model (likelihood), and $f_{X|Y}(x \mid y)$ is the posterior distribution used by the receiver to make Maximum A Posteriori (MAP) symbol decisions.

Summary Matrix: Comprehensive Comparison of Bivariate Operations

Operation / Concept	Mathematical Governing Expression	Independence Simplification	Physical Meaning / Role
Marginal Density	$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y)\,dy$	$f_X(x)\int f_Y(y)dy = f_X(x)$	Complete projection onto X-axis.
Conditional Density	$f_{X\|Y}(x \mid y) = \frac{f_{XY}(x,y)}{f_Y(y)}$	$f_{X\|Y}(x \mid y) = f_X(x)$	Updated density given knowledge $Y = y$.
Sum of RVs	$Z = X + Y \implies f_Z(z) = \int f_{XY}(w, z-w)dw$	**Convolution:** $f_Z = f_X * f_Y$	Total noise from independent sources.
Quotient of RVs	$Z = X/Y \implies f_Z(z) = \int \lvert w\rvert f_{XY}(zw,w)dw$	$\int \lvert w\rvert f_X(zw)f_Y(w)dw$	Signal-to-noise ratio; slope of vector.
Cauchy Distribution	$Z = X/Y$ **for** $X, Y \sim \mathcal{N}(0,1) \implies \frac{1}{\pi(1+z^2)}$	**Quotient of two standard Gaussians**	Heavy-tailed ratio; undefined moments.
Continuous Bayes	$f_{X\|Y}(x \mid y) = \frac{f_X(x)f_{Y\|X}(y\|x)}{\int f_X(u)f_{Y\|X}(y\|u)du}$	$f_{X\|Y}(x \mid y) = f_X(x)$	Optimal Bayesian estimation / inference.

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